Monday, November 12, 2018

IT 540 Weekly Content

Week 11

- **Final Project Objectives:**

  - The final project is intended to mimic what will be expected of you in a work-related analytic project. In a work project, people are often tasked by management with a question to answer from a dataset. Therefore, they obtain the data, use it for analysis, and then make a presentation to management to explain their results. Those results may or may not be what management was hoping to hear, but it is important to present the results demanded by the data. To this end, you must be confident that you performed a correct analysis *or* that your analysis was actually lacking, but you understand how it was lacking, and you have a plan to address the shortcomings in a future analysis.

- **Guidelines:**

  1. Select a dataset of your choice to analyze. Do not spend a large amount of time looking for an appropriate dataset. *If you want ME to select one, tell me **quickly** and I will provide it, or tell you where to find it.* Ideally, your data would be something that you created yourself or that you otherwise own, e.g. data captured from an exercise monitoring device, GPS data from a hiking or biking app, etc. Please do not use data from your place of employment, so we can avoid any questions about proprietary or sensitive data. If you use personal information, I will not share anything I learn from your data or from your analysis with anyone. If you prefer, you may use data from any publicly available and legal source, e.g. any dataset you find from www.data.gov, or any of the datasets that are available from sources we have seen or that you are otherwise aware of. (See my handout from Week 0 with many sources of data.) The reason I suggest personal data is simply because you will take a greater interest in it, which will motivate you to perform excellent work. So if you choose an outside source instead, try to find something that you find truly interesting.
  2. You have a lot of freedom in the exact form of your analysis, but you must address each of the "5 core activities of data analysis", as described on page 7 of Peng and Matsui, and further elaborated in that text. For example, the first core activity of determining a question to answer is further refined in chapter 2 as being one of 6 types of questions. The core activities are:

A. Stating and refining the question.

B. Exploring the data.

C. Building formal statistical models.

D. Interpreting the results.

E. Communicating the results.

- **Grading Criteria:** 280 points (28%) out of a course total of 1,000 points
  - *Selection of data to analyze: 0 points.* This is the starting point.
  - *Part A: Stating and refining the question: 20 points*. You will turn in a written paper (see Part E below) where you will describe this question. It is perfectly acceptable for you to state the question once and not revise it during the course of your analysis. However, if you do refine or revise the initial question, please include a brief description/motivation for why you decided to change it. Whatever question or hypothesis you present, it should be a question that is answerable in some quantitative or analytical way with the data you have selected. (But see below for a possible exception to this.)
  - *Part B: Exploring the data: 35 points*. You should write R code to perform basic data exploration, including any needed data cleansing as well as creating useful charts/graphs to help you understand the nature of your data, helping you either to validate or refine your initial question.
  - *Part C: Building formal statistical models: 75 points*. Your model may be as simple as a linear regression or as complex as a neural network, or other advanced machine learning model. However, if you state an obvious, very easy question and answer it with a simple linear model of one variable, do not expect full credit. More significant work to answer more significant questions will be rewarded in this class, just as it is in the workplace.
  - *Part D, Interpreting the results: 75 points*. In your written paper, describe whether your analysis answered or did not answer the question from Part A, and why or why not. Include copies of any charts or graphs created by your code.
  - *Part E: Communicating the results: 75 points*. Submit a written document in MS Word, Mac Keynote, or PDF which:
    - contains a brief description of your dataset
    - describes the question your analysis is attempting to answer in Part A

- describes your exploratory analysis (Part B) and your formal model (Part C), e.g. logistic regression, K-nearest neighbor, etc., and *why* you chose to use this model instead of something else.
- Your interpretation of the results of your analysis (Part D). (Did you answer the question? Fail to answer it? Why or why not?)
- Conclusions: If you answered the question successfully, describe what you can do with this information? Why is it useful or valuable? If you failed to answer the question, describe why, and whether or not it might be possible to perform a new analysis to try again.

- **DELIVERABLES: Here are the deliverables that you must submit:**
  - If you use your own data (not a dataset that I have given you), then you MUST give me a copy of it or tell me where to find it for myself. **Failure to share your data with me, will cost you 15 points!**
  - Submit a copy of your R code file.
  - Submit your written report that is described above in Part E.